LING406_Lec14_LangDevelopment_Notes

Course: LING406: Psycholinguistics
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Section 1: Nature and Nurture in Child Language

- **Nativist** positions on language acquisition come from **Chomsky's critique of Skinner**. Skinner's behaviorist approach explained behavior, including language, through general learning mechanisms such as reward and punishment. Chomsky argued that this kind of explanation cannot account for what children actually accomplish when they learn a language.
- For Chomsky, the most important feature of language is its **generativity**. A speaker can produce and understand an unlimited number of sentences they have never heard before, so a language cannot be a stored list of sentences. Under this view, **acquiring a language means identifying the abstract grammar** that produces the sentences of that language. The child's job is to work out the hidden system of rules, not to memorize the sentences they happen to hear.
- Chomsky argued that the general learning mechanisms studied by behaviorists were not up to this task, and he gave two main reasons.
    - **Grammatical and ungrammatical sentences are not "similar" in any straightforward way.** A grammatical sentence and an ungrammatical one can differ by a single word, while two grammatical sentences can look completely different from each other. A learner that groups things by surface similarity therefore has no reliable way to sort sentences that follow the grammar from sentences that break it.
    - **The context for language acquisition does not look like operant conditioning.** In operant conditioning, a behavior becomes more or less frequent depending on whether it is followed by reward or punishment. Children are not consistently rewarded for grammatical sentences and punished for ungrammatical ones, so the reward and punishment contingencies that operant conditioning depends on do not hold.
- Instead of general learning, nativist approaches assert two things that the child brings to the task from birth.
    - The first is **innate knowledge of what language is like**, called **Universal Grammar**. This is built-in knowledge of the general form that any human language can take, which narrows down the possible grammars a child has to consider.
    - The second is **language-specific acquisition processes**. These are learning processes dedicated to language alone, separate from the general mechanisms used to learn everything else.
- Chomsky supported this view with an argument about speed and uniformity. All normal children acquire essentially comparable grammars, and they do it with remarkable rapidity. He took this as evidence that humans are somehow specially equipped with a **data-handling or "hypothesis-formulating" ability** of unknown character, meaning a built-in capacity for forming and testing guesses about the grammar of the language around them.

Section 2: Categorical Perception

- **Categorical perception** happens when listeners hear sounds that vary smoothly and continuously in their physical properties as falling into a small number of separate categories. Listeners are much better at telling apart two sounds from different categories than two sounds from the same category, even when the physical difference between the two pairs is exactly the same size.
- The classic stimuli are a set of 14 synthetic speech syllables, numbered 1 through 14. Each syllable is built from two bands of sound energy called **formants**, plotted as frequency in cycles per second (cps) over time in milliseconds, with each syllable lasting roughly 300 ms.
    - The lower formant, the **first formant**, is the same in all 14 stimuli. It starts low and rises quickly to a steady level of a few hundred cps.
    - The upper formant, the **second formant**, is the only thing that changes across the continuum, and the change is in its opening movement, called the **transition**. In stimulus 1 the second formant starts well below its steady level and rises steeply up to about 2200 cps. Moving through the continuum, this rise gets smaller, it becomes nearly flat around stimuli 6 to 8, and from stimulus 9 onward the transition starts above the steady level and falls, with stimulus 14 having the steepest fall.
    - The result is a single physical dimension, the direction and size of the second-formant transition, that changes in equal steps from one stimulus to the next.
- **Labeling data** come from asking listeners to identify each stimulus as b, d, or g.
    - Stimuli 1 through 3 are labeled b nearly 100% of the time, and b responses drop sharply around stimulus 4.
    - Stimuli 5 through about 8 are labeled d nearly 100% of the time, and d responses then drop sharply around stimuli 9 and 10.
    - Stimuli 11 through 14 are labeled g nearly 100% of the time.
    - The switch between categories is abrupt rather than gradual. Even though the stimuli change in equal steps, listeners hear a long stretch of b, a sudden change, a long stretch of d, another sudden change, and then g.
- **Discrimination data** come from asking listeners to tell apart two stimuli that are a fixed number of steps apart on the continuum: one step, two steps, or three steps. Performance is plotted as percent correct against the value of the "A" stimulus, which is the first member of each pair, with 50% correct being chance.
    - In **one-step discrimination**, listeners are near chance for most pairs but rise to about 80% correct for pairs near stimuli 4 and 5 and to about 75% for pairs near stimulus 8. These are exactly the places where the labeling curves cross from b to d and from d to g.
    - In **two-step discrimination**, the peaks are higher, reaching close to 100% for pairs starting around stimulus 3 and around stimuli 7 and 8, while pairs that fall inside a category, such as those starting at stimuli 5 and 6, drop to about 50%.
    - In **three-step discrimination**, performance is again near 100% for pairs that cross a category boundary and drops toward 50% to 75% for pairs inside a category.
    - Larger step sizes make every pair physically more different, which raises discrimination overall, but the peaks still sit at the category boundaries and the dips still sit inside the categories.
- Each discrimination graph compares **obtained** performance with **predicted** performance. The predicted values are calculated from the labeling data, under the assumption that listeners can only tell two sounds apart if they give them different labels. The obtained curves follow the same pattern of peaks and dips as the predicted curves, sometimes running somewhat higher, which means listeners' ability to discriminate the sounds is closely tied to how they categorize them.

- **Early theories treated categorical perception as both language-specific and species-specific.** At the time, only humans were known to show categorical perception, and they were only known to show it for speech sounds. This made it look like a special property of a uniquely human language system.
- Categorical perception is a compelling example of **similarity between stimuli being defined in linguistic rather than physical terms**. Physically, stimulus 3 and stimulus 5 are just as far apart as stimulus 6 and stimulus 8. Listeners treat the first pair as clearly different and the second pair as nearly identical, since 3 and 5 belong to different speech categories while 6 and 8 belong to the same one.

Section 3: Methods for Testing Categorical Perception in Infants

- Infants cannot label sounds or say whether two sounds are the same or different, so researchers measure discrimination through behaviors infants already produce. Two methods are central here.
- The **conditioned head turn procedure**, used in Janet Werker's infant studies lab (infantstudies.psych.ubc.ca), relies on training an infant to turn toward a reward when a sound changes.
    - The infant sits on a parent's lap facing an experimenter across a small table, and the experimenter keeps the infant's attention forward with toys. Beside the infant is a box that can light up and reveal a moving toy, such as a stuffed rabbit.
    - The infant hears a repeating sound, and when the sound changes to a different one, the toy lights up. Over time the infant learns that a change in sound predicts the toy, so the infant starts turning toward the box as soon as the sound changes, even before the toy appears.
    - Once this conditioning is in place, a head turn after a sound change indicates that the infant detected the change. Both the parent and the experimenter wear headphones, so they cannot hear the sounds and cannot accidentally signal the infant when a change happens.
- **High amplitude sucking** measures how strongly and how often an infant sucks on a pacifier. The pacifier is connected to a device that records each suck as a pressure peak over time, producing a trace of sucking activity. The apparatus appears in Rochat & Striano (2001, IBD), a paper unrelated to speech perception, but the recording method is the same.

Section 4: Eimas et al. (1971): Categorical Perception in Infants

- **Eimas et al. (1971, Science)** used high amplitude sucking to test whether infants perceive speech categorically. Each strong suck caused a speech syllable to play, so infants controlled how often they heard the sound.
    - When a sound is new, infants suck more to keep hearing it. As they get used to the sound, their interest fades and their sucking rate drops, a decline called **habituation**.
    - After the sucking rate dropped, the experimenters switched to a new syllable, a moment called the **stimulus shift**. If the infant can tell the new syllable apart from the old one, sucking should increase again. If the infant hears the new syllable as the same as the old one, sucking should keep declining.
- The syllables differed in **voice onset time (VOT)**, which is the time between the release of a consonant and the start of vocal fold vibration, or voicing. VOT is measured in milliseconds, with positive values meaning voicing starts after the release and negative values meaning voicing starts before it.
    - In a syllable with a VOT of $+10$ msec, the three formants, labeled F_1, F_2, and F_3 and plotted in kilocycles (thousands of cycles per second), all begin at nearly the same moment. F_1 begins just 10 ms after F_2 and F_3.

    - In a syllable with a VOT of $+100$ msec, F_1 does not begin until 100 ms after the start of the syllable. During that delay, F_2 and F_3 contain noisy energy rather than clear voicing, and only after 100 ms do all three formants carry voicing together.
- The main experiment had three conditions, run at each of two age levels, and the results described here come from the **4-month-old infants**.
    - In condition **20D**, the two syllables differed by 20 ms of VOT and came from **different adult phonemic categories**. The stimuli had VOT values of $+20$ and $+40$ msec, which fall on opposite sides of the adult category boundary.
    - In condition **20S**, the two syllables again differed by 20 ms of VOT, but both came from the **same adult phonemic category**. The stimuli had VOT values of -20 and 0 msec, or $+60$ and $+80$ msec.
    - Condition **0** was a **control** in which the stimulus never changed. For this group, the "shift" marks only the point in time when a change would have happened.
- Sucking was measured as the mean number of sucking responses per minute, starting from a **baseline** rate labeled B, then over the 5 minutes before the shift and the 4 minutes after it.
    - In the **20D** condition, sucking rose from a baseline of about 26 to about 52 to 55 responses, then declined to about 39 and 34 in the 2 minutes before the shift. Right after the shift, sucking jumped back up to about 51 and stayed between about 46 and 48 for the remaining minutes.
    - In the **20S** condition, sucking rose from a baseline of about 25 to a peak of about 66, then fell to about 43 and 35 before the shift. After the shift, sucking stayed low at about 38 and kept falling to about 28.
    - In the **0** control condition, sucking rose from a baseline of about 24 to about 58, fell to about 36 and 30 before the shift, and continued falling afterward to about 16.
- In both 20D and 20S, the physical change at the shift was the same size, 20 ms of VOT. Infants recovered their sucking only when the change crossed the adult category boundary. When the change stayed inside one category, infants behaved almost like the control group, which heard no change at all. This is the same pattern as adult categorical perception, with good discrimination between categories and poor discrimination within a category, appearing in infants only 4 months old.

Section 5: Why Infant Categorical Perception Seemed Like Evidence for Innateness

- **Categorical perception was considered the first step in turning speech into a symbolic format** that later linguistic processes could use. Speech arrives as a continuously varying sound signal, but processes such as recognizing words and building sentences need discrete units. Categorical perception sorts the continuous signal into those discrete categories.
- Four-month-olds are very young, and their other abilities are still limited.
    - Their gross motor milestones at this age include manipulating objects with two hands and holding their head and chest up while lying on their stomach, a position called **prone**.
    - Their color vision and general visual acuity are still developing.
    - They are just starting to produce vowel-like sounds instead of creaky whines.
- An apparently language-specific ability appearing this early in development, before infants can even produce speech-like vowels, seemed like strong **evidence for innateness**. The infants had very little time to learn speech categories from experience, so it seemed the categories must have been built in from the start.

Section 6: Perceptual Narrowing for Speech Categories

- **Perceptual narrowing** is the decline, over the first year of life, in infants' ability to discriminate speech contrasts that are not used in their own language. The key evidence comes from **Werker & Tees (1984a, "Cross-language speech perception: Evidence for perceptual reorganization during the first year of life," Infant Behavior and Development, 7, 49–63)**.
- English-learning infants were tested on two **non-English contrasts**, which are pairs of sounds that are distinct speech sounds in another language but not in English.
    - The **Hindi Ta/ta** contrast consists of two t-like sounds that Hindi treats as different, produced with the tongue touching different places in the mouth.
    - The **Salish ki/qi** contrast consists of two k-like sounds produced at slightly different places at the back of the mouth, which Salish treats as different.
    - English speakers hear each pair as a single English sound, a t in the first case and a k in the second.
- Performance was measured as the **percentage of infants at each age who reached a discrimination criterion**, meaning a set level of reliable performance.
    - Among English-learning infants aged **6 to 8 months**, about 95% reached criterion on the Hindi contrast and about 80% on the Salish contrast.
    - Among English-learning infants aged **8 to 10 months**, these numbers dropped to about 67% for Hindi and about 52% for Salish.
    - Among English-learning infants aged **10 to 12 months**, only about 20% reached criterion for Hindi and about 10% for Salish.
    - **Hindi-learning and Salish-learning infants**, who hear these contrasts in their own languages, reached criterion at close to 100% on their native contrasts.
- At around six months, English-learning infants show evidence of discriminating speech contrasts they have never heard before. Over the course of development, the ability to perceive these contrasts declines, and by the end of the first year most English-learning infants no longer discriminate them, while infants exposed to the languages that use these contrasts keep the ability.
- Perceptual narrowing connects directly to categorical perception. For an English-learning infant at 10 to 12 months, the two Hindi t sounds have come to fall inside a single English category. A difference inside one category is exactly the kind of difference that categorical perception makes hard to detect, so the once-discriminable contrast now behaves like a within-category difference.

Section 7: The Nativist Narrative of Speech Development

- The findings on infant categorical perception and perceptual narrowing fit together into a very compelling nativist narrative.
    - First, speech perception depends on **categorical perception abilities that are a special property of the language system**, not part of general hearing.
    - Second, these abilities are **present very early in development**, already in place by 4 months.
    - Third, children start off as **"citizens of the world,"** able to discriminate all speech contrasts used in any language. Under this view, infants have innate, language-general knowledge about speech, and they then rapidly tune their perceptual systems to the specific language in their environment.
- In this narrative, **the role of learning is selection rather than construction**. The full set of possible phonetic contrasts is innately specified, and experience simply selects among them, narrowing the inventory down to just the contrasts that are meaningful in the child's language. Contrasts that the child's language does not use are lost, as in the Werker & Tees data.

Section 8: Problems for the Nativist View

- Several findings create problems for this narrative, some about categorical perception itself and some about infant speech perception.
- **Categorical perception is not limited to language.** Listeners also show categorical perception for sounds that are not speech, which weakens the claim that it is a special property of the language system.
- **Categorical perception is not limited to humans.** Non-human animals can also show categorical perception of speech sounds, which weakens the claim that it is a uniquely human, species-specific ability.
- **Categorical perception is task-dependent.** Whether listeners show sharp category boundaries depends on the details of how they are tested, which suggests it is not a fixed, built-in property of perception.
- **Infant speech perception is not actually "universal."** Infants do not start out able to discriminate every possible speech contrast, which contradicts the idea that they begin as "citizens of the world" with all contrasts innately available.

Section 9: Categorical Perception of Non-Speech Sounds in Adults

- **Cutting & Rosner (1974, Perception & Psychophysics)** tested whether adults show categorical perception for sounds that are not speech. The stimuli varied in **rise time**, which is how long a sound takes to grow from silence to its full loudness.
- Three types of stimuli were used across Experiments I and II, each at rise times ranging from short to long, with the sounds plotted as **oscillograms**, records of the sound wave's amplitude over time, each lasting about 400 ms.
    - **Sawtooth waves** are buzzy, musical-sounding non-speech tones. At a 10 msec rise time the wave reaches full amplitude almost instantly, and at a 70 msec rise time it swells gradually before reaching full amplitude.
    - **Speech syllables** were also made with short and long rise times, with the 10 msec version starting abruptly and the 70 msec version starting more gradually.
    - **Sine waves** are pure tones, also made in 10 msec and 70 msec versions, with the same contrast between an abrupt and a gradual onset.
- For the sawtooth waves, rise time varied from 0 to 80 ms in 10 ms steps. Listeners labeled each sound as either a **pluck**, like a plucked string, or a **bow**, like a string played with a bow.
    - **Pluck responses** were near 100% at rise times of 0 and 10 ms, about 88% at 20 ms, and about 80% at 30 ms. They then dropped to about 50% at 40 ms, about 25% at 50 ms, and down to only a few percent at 80 ms.
    - **Bow responses** mirrored this pattern, staying near 0% at 0 and 10 ms, rising to about 50% at 40 ms, and reaching about 93% at 80 ms.
    - The **category boundary** between pluck and bow therefore sits at a rise time of about 40 ms, where listeners give each label about half the time.
- **Discrimination** was tested on pairs of sawtooth waves that differed by 20 ms in rise time, such as 0 versus 20 ms, 10 versus 30 ms, and so on up to 60 versus 80 ms.
    - Pairs inside a category, such as 0 versus 20 ms or 50 versus 70 ms, were discriminated at only about 55% to 59% correct.
    - Pairs near the boundary were discriminated much better, with 20 versus 40 ms at about 72% and **30 versus 50 ms**, the pair that straddles the 40 ms boundary most evenly, peaking at about 77% correct.
- This is the full categorical perception pattern, with an abrupt shift in labeling and a discrimination peak at the category boundary, produced by sounds that are not speech at all. Categorical perception in adults is therefore not limited to language.

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Section 10: Categorical Perception of Non-Speech Sounds in Infants

- **Jusczyk et al. (1977, Perception & Psychophysics)** used the high amplitude sucking method from Eimas et al. to test whether infants also perceive non-speech sounds categorically. Infants sucked to hear a sound, their sucking declined as they habituated, and then the sound changed at the stimulus shift.
- There were three conditions.
    - In the **no change** condition, the sound never changed, so this group served as the control.
    - In the **within categories** condition, the sound changed to a different stimulus from the same category.
    - In the **between categories** condition, the sound changed to a stimulus from a different category.
- Sucking was measured as the mean number of sucking responses per minute, starting from a baseline rate labeled B, then over the 3 minutes before the shift and the 4 minutes after it.
    - In the **no change** condition, sucking started at a baseline of about 15, rose to about 20, and fell to about 13 and 12 before the shift. After the shift, sucking stayed flat at about 12 for all 4 minutes.
    - In the **within categories** condition, sucking started at about 14, rose to about 20, and fell to about 12 before the shift. After the shift, it stayed at about 11 to 12, matching the no change group.
    - In the **between categories** condition, sucking started at about 13, rose to about 20, and fell to about 13 and 11 before the shift. After the shift, sucking jumped to about 18 and then 16 before settling back to about 12.
- Infants recovered their sucking only when the non-speech sound changed across a category boundary, and a within-category change produced no more sucking than no change at all. Infants therefore show the same categorical pattern for non-speech sounds that Eimas et al. found for speech, which undermines the claim that infant categorical perception is a special property of the language system.

Section 11: Categorical Perception of Speech Sounds by Chinchillas

- **Ohlemiller et al. (1999, Behavioral Brain Research)**, building on earlier work by **Kuhl & Miller (1975)**, tested whether chinchillas perceive human speech sounds categorically. The stimuli came from a **/ga/–/ka/ continuum** varying in VOT. A /ga/ has a short VOT, with voicing starting soon after the consonant release, and a /ka/ has a long VOT, with a longer delay before voicing.
- The chinchillas were trained using **operant conditioning**, in which either the 0-lag stimulus or the 80 ms-lag stimulus served as an S^D **for food**.
    - An S^D, or **discriminative stimulus**, is a signal that tells the animal a response will now be rewarded. The chinchilla learns to respond when it hears the S^D and not respond when it hears the other endpoint.
    - The animals were trained only on the two endpoints, 0 ms and 80 ms. They were then tested on the stimuli in between, and their responses indicate whether each in-between stimulus was treated like the /ga/ endpoint or the /ka/ endpoint.
- Results are plotted as the **proportion of trials labeled /ga/** across VOT values from 0 to 120 ms, for two chinchillas (C1 and C3) and two human listeners (S3 and S4).
    - Chinchilla **C1** labeled short-VOT stimuli as /ga/ about 90% of the time up to about 30 ms. Its /ga/ responses then fell steeply, crossing 50% at a VOT of roughly 45 ms and dropping to about 15% to 20% by 70 to 80 ms.
    - Chinchilla **C3** labeled short-VOT stimuli as /ga/ more than 90% of the time up to about 30 ms, and its labeling crossed 50% at roughly 50 to 55 ms.

    - Human **S3** labeled stimuli as /ga/ nearly 100% of the time up to about 45 ms, with a steep drop and a boundary at roughly 60 to 65 ms.
    - Human **S4** labeled stimuli as /ga/ nearly 100% of the time up to about 30 ms, with a very steep drop and a boundary at roughly 35 to 40 ms.
- Both chinchillas and humans produce the same type of labeling curve, with a long flat stretch of /ga/ responses, a sharp transition, and a long flat stretch of /ka/ responses. The chinchillas' boundaries fall within the range of the human boundaries. Categorical perception of speech sounds is therefore not limited to humans, and an animal with no language can divide the VOT continuum into two categories in much the same place as a human listener.
- The location and steepness of the boundary changed with the details of the stimuli, in both species.
    - Three stimulus sets were tested: a continuum running from **0 to 80 ms**, a continuum running from **0 to 120 ms**, and an **enhanced burst** continuum.
    - For chinchilla C3 and human S3, the 0 to 120 ms continuum moved the boundary to longer VOTs than the 0 to 80 ms continuum, and for C3 it also made the curve less steep. The same VOT value could therefore be labeled differently depending on the range of stimuli it was presented with.
    - In the enhanced burst stimuli, the experimenters increased the intensity of the **frication burst**, the short noise at the consonant release, and the **aspiration noise**, the breathy noise before voicing begins. This shifted the category boundary to a shorter VOT, made the curve steeper, or both, in one chinchilla (C3) and one human listener (S3).
    - Changes in the stimulus set and in acoustic details other than VOT moving the boundary is consistent with categorical perception being **task-dependent** rather than a fixed property of perception.

Section 12: Some Contrasts Are Less Salient Than Others for Infants

- **Narayan et al. (2010)** tested whether infants really start out able to discriminate every speech contrast. They compared two contrasts between **nasal consonants**, sounds made with air flowing through the nose, such as m, n, and the ng sound written ŋ.
    - The **[ma]–[na]** contrast is native to English, meaning English uses it to distinguish words.
    - The **[na]–[ŋa]** contrast is non-native for English-learning infants. Both are a **place contrast**, a difference in where in the mouth the consonant is made, and in [na]–[ŋa] the nasal appears in the **onset**, the beginning of the syllable, a position where English never uses ŋ.
- English-hearing infants at **4 to 5 months** were tested on **same trials**, where one syllable repeated, and **change trials**, where the syllables switched. Looking time was measured in seconds, with longer looking on change trials than on same trials indicating that infants noticed the change. Error bars represent standard error, a measure of how much the averages would be expected to vary.
    - For **[ma]–[na]**, looking time rose from about 3.4 seconds on same trials to about 5.4 seconds on change trials, so infants discriminated this contrast.
    - For **[na]–[ŋa]**, looking time was about 4.5 seconds on same trials and about 4.2 seconds on change trials, so infants did not look longer when the sound changed and showed no evidence of discrimination.
- The **/na/–/ŋa/ place contrast for nasals in onset is not readily perceived by four-month-olds, independent of language experience**, meaning young infants have trouble with it even when their language uses it. In contrast, **/ma/–/na/ is relatively salient**, meaning it stands out and is easy for infants to detect. Infants therefore do not start out with equal access to all speech contrasts.

Section 13: Experience Increases the Salience of a Hard Contrast

- Narayan et al. (2010) also tested **Filipino-learning infants** on the [na]–[ŋa] contrast, which is native to Filipino, at two ages.
    - At **6 to 8 months**, Filipino-learning infants looked about 3.8 seconds on both same trials and change trials, so they did not yet discriminate the contrast even though it occurs in their language.
    - At **10 to 12 months**, looking time rose from about 3.3 seconds on same trials to about 4.8 seconds on change trials, so these older infants did discriminate it.
- **Between 6 and 10 months, exposure to this contrast in a Filipino language environment increases its salience.** Infants in this environment become able to detect a contrast that they could not detect earlier.
- **Not all speech contrasts are available innately.** The "citizens of the world" narrative says infants begin with every contrast and experience only removes the ones their language does not use. The [na]–[ŋa] contrast follows the opposite pattern, starting out hard to perceive and becoming perceivable only after months of hearing it. **Exposure to contrasts in the environment is necessary for acquisition in some cases**, so learning has to build some contrasts, not just select among innate ones.

Section 14: The Emergentist Alternative

- **Emergentist approaches** are the main alternative to nativism. They assume that **associative learning mechanisms are sensitive to statistical regularities in the environment**, so language structure emerges from general learning applied to the patterns in the input, rather than from innate language-specific knowledge.
- The basic associative mechanism is the strengthening of a connection between neurons that are active together, as in classical conditioning of salivation to a bell.
    - A **food detector** neuron connects to a **salivation** neuron, so food on its own triggers salivation from the start.
    - A **bell** neuron also connects to the salivation neuron, but at first this connection is too weak to trigger salivation by itself.
    - When the bell sounds at the same time food is presented, the bell neuron is active at the same moment the salivation neuron is being driven by the food detector. This joint activity strengthens the **synapse** between the bell neuron and the salivation neuron.
    - After repeated pairings, that synapse becomes strong enough that the bell alone activates the salivation neuron, and the animal salivates to the bell.
- The emergentist challenge is to extend this kind of associative learning beyond simple **conditioning** to support learning for **pattern recognition**, such as recognizing the speech categories of a language from the sounds a child hears.

Section 15: Learning the Statistics of Vowel Distributions

- When **Kuhl** and others talk about **learning the statistics of distributions in the environment**, they mean learning the shapes of distributions like those of the American English vowels in a **vowel space**.
- A **vowel space** is a two-dimensional map defined by the **first formant** (F_1) and the **second formant** (F_2), the two lowest bands of energy in a vowel. F_1 runs along the horizontal axis from about 300 to 1200 Hz, F_2 runs along the vertical axis from about 600 to 3400 Hz, and every vowel token a speaker produces lands at one point in this space.
    - Tokens of each vowel form a cloud of points, and an ellipse drawn around each cloud marks the region where most tokens of that vowel fall.
    - The vowel labeled **i** sits in the upper left, with low F_1 of about 300 to 500 Hz and very high F_2 of about 2200 to 3400 Hz. The vowel labeled **u** sits in the lower left, with similarly low F_1 but much lower F_2 of about 850 to 1600 Hz. The vowel

labeled **a** extends to the far right, with high F_1 of about 650 to 1200 Hz. The vowel labeled **ae** sits in the upper middle, with F_1 of about 550 to 850 Hz and F_2 of about 1900 to 2700 Hz.
    - The clouds are spread out, and many of the ellipses overlap heavily, especially among the vowels in the middle and lower left of the space.
- This overlap is why **speech categorization is difficult**. Different speakers and different productions of the same vowel vary widely, and a token at a given combination of F_1 and F_2 could belong to more than one vowel category, so no single pair of formant values defines a vowel.
- One possibility is that we **learn the vowels of our language by extracting the statistics of these distributions**. A listener who hears thousands of vowel tokens could pick up where the tokens cluster and how widely they spread, without any innate knowledge of the vowel categories.
- Under this view, a **"prototypical" /a/** is just the **mean**, or possibly the **mode**, of the distribution of all the /a/ sounds we hear. The mean would be the average F_1 and average F_2 across all /a/ tokens, and the mode would be the combination of F_1 and F_2 that occurs most often.

Section 16: Maye et al. (2002): Infants Learn from Distributions

- **Maye et al. (2002, Cognition, 82, B101–B111)** tested whether infants actually learn categories from the statistics of what they hear. Infants **listened passively**, with no reward or feedback, to one of two distributions of **voice onset time** along a continuum of eight [da]–[ta] stimuli. Token 1 was the endpoint [da] and Token 8 was the endpoint [ta].
- During the **familiarization phase**, the two groups heard the same eight tokens, but with different frequencies.
    - The **bimodal** group heard Tokens 1 through 8 presented 4, 16, 8, 4, 4, 8, 16, and 4 times. The frequencies form two peaks, at Token 2 and Token 7, which provides evidence for **two different categories**, one near the [da] end and one near the [ta] end.
    - The **unimodal** group heard Tokens 1 through 8 presented 4, 4, 8, 16, 16, 8, 4, and 4 times. The frequencies form a single peak in the middle, at Tokens 4 and 5, which provides evidence for **a single category**.
    - Both groups heard 64 tokens in total, and both heard Token 3 and Token 6 exactly 8 times each. The only difference between the groups was the overall shape of the distribution.
- **At test**, all infants heard stimuli **3 and 6** in two kinds of trials. In **alternating trials** the two stimuli switched back and forth, and in **non-alternating trials** a single stimulus repeated. Infants who hear 3 and 6 as different sounds should look differently on the two trial types, while infants who hear them as the same sound should treat both trial types alike.
- Mean looking times in seconds, with standard errors in parentheses, were as follows.
    - **6-month unimodal**: 4.85 (0.47) on alternating trials and 4.53 (0.51) on non-alternating trials.
    - **8-month unimodal**: 4.98 (0.63) on alternating trials and 5.20 (0.56) on non-alternating trials.
    - **6-month bimodal**: 5.66 (0.44) on alternating trials and 6.41 (0.32) on non-alternating trials.
    - **8-month bimodal**: 5.45 (0.52) on alternating trials and 6.15 (0.56) on non-alternating trials.
- **Only infants pre-exposed to the bimodal distribution were sensitive to the difference between conditions.** Bimodal infants at both ages looked about 0.7 seconds longer on non-alternating trials than on alternating trials. Unimodal infants showed only small differences that went in opposite directions at the two ages.

- The unimodal infants behaved **as if they had learned to treat stimuli 3 and 6 as members of the same category**. Both groups heard stimuli 3 and 6 the same number of times, so the difference came entirely from the shape of the distribution. A few minutes of passive listening to a one-peak or two-peak distribution was enough to change how infants perceived the same pair of sounds, which is the kind of statistical learning emergentist approaches propose.

Section 17: The Social and Interactive Context for Learning

- Experimental evidence indicates that **social interaction is important for learning** speech sounds, beyond simple exposure to the sounds themselves.
- In a **foreign-language exposure** study, **nine-month-old American infants** were exposed to **Mandarin Chinese** in **twelve 25-minute sessions**.
    - One group received **live exposure**, with a Mandarin speaker playing and reading books with the infants in person. Other groups received the same Mandarin material through **television**, as audiovisual recordings, or through **audio only**.
    - A control group received live sessions in **English** instead of Mandarin.
    - After the exposure period, all infants were tested on a **Mandarin phonetic contrast** using the **head-turn procedure**, with 50% correct as chance.
- Infants learned the Mandarin contrast only from live exposure.
    - Infants with **live Mandarin exposure** scored about 65% to 66% correct, compared with about 56% for the **English control** group.
    - Infants with **TV exposure** and **audio-only exposure** each scored about 56%, no better than the English controls.
    - **The same exposure to a Mandarin speaker is more effective for learning when it happens in person and is interactive than when it is viewed as a video.**
- In a **social response manipulation** study, **eight-month-old infants** received either contingent or non-contingent social feedback from their mothers in response to their vocalizations.
    - In the **contingent condition (CC)**, Visit 1 was a 30-minute familiarization session. Visit 2 had a 10-minute baseline, then 10 minutes in which the mother responded socially right after the infant's vocalizations, then a 10-minute **extinction** period in which these responses were removed.
    - In the **yoked-control condition (YC)**, the schedule was the same, but the mother's social responses during the middle 10 minutes were **non-contingent**. The infant received the same kind of social attention, but its timing was not linked to the infant's own vocalizations.
- Learning was measured as the **proportion of syllables to vocalizations**, which captures how often the infant's vocalizations contained more mature, syllable-like sounds.
    - At **baseline**, the proportion was about 0.08 in the contingent group and about 0.04 in the non-contingent group.
    - During the **social response** period, the contingent group rose to about 0.32, while the non-contingent group stayed at about 0.05.
    - During **extinction**, the contingent group dropped to about 0.20, still well above its baseline, while the non-contingent group was about 0.09.
    - **Contingent social feedback increased the quantity and complexity of infants' vocalizations**, and non-contingent feedback of the same amount did not.
- **Social interactions contingent on the production of different sounds seem to function much like a reinforcing stimulus.** The mother's response, delivered right after a vocalization, increased the infant's production of syllable-like sounds the way a reward

increases a behavior in operant conditioning, and the effect partly faded once the responses stopped.

- There are two ideas about why social interaction matters.
  - One idea is that **humans find social interaction with other humans innately reinforcing**, so social responses act as a built-in reward that drives learning.
  - Another idea is that **humans are innately good at using social cues** to learn from other people, such as following where another person is attending.

Section 18: Statistical Learning with Constraints

- The emergentist view does not claim that infants learn from raw statistics with no limits. Statistical learning operates with **constraints**, and "constraints" refers to several different things.
  1. **What stimulus properties can be contrastive.** Some properties of speech never distinguish one speech category from another, such as who is speaking or how fast they are speaking. A /da/ is still a /da/ whether a child or an adult says it, quickly or slowly, so the learner must not form categories along these dimensions even though they vary a great deal in the input.
  2. **Constraints based on earlier learning.** What an infant has already learned shapes what it can learn next, so later statistical learning is limited and guided by the categories and patterns acquired earlier.
  3. **Social and interactive contexts for learning.** The live Mandarin exposure and contingent feedback findings indicate that the same input is learned more effectively, or only learned at all, when it comes through social interaction with another person.